

EXAM 7 – SPRING 2012

5. (2.25 points)

A loss reserve actuary has reviewed three cumulative paid loss triangles to test whether the assumptions underlying the chain ladder method are met.

a. (0.75 point)

State the three chain ladder assumptions as described by Mack.

b. (1.5 points)

For each of the following situations, discuss whether any of Mack's assumptions are violated.

- i. The first triangle shows a faster claims settlement pattern in the most recent calendar year resulting from the use of new claims management software.
- ii. For the second triangle, the actuary found that the most appropriate selection method for loss development factors was to use an all year volume-weighted average approach.
- iii. In the third triangle, the 36-48 month loss development factors are inversely proportional to the 24-36 month loss development factors. These relationships do not appear to be random.

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Question 5 Sample Answer

Solution 1

a)

1. $E(C_{i,K} | C_{i,K-1}, \dots, C_{i,0}) = f_K \times (C_{i,K-1} + C_{i,K-2} + \dots)$
Incremental losses equal a loss developmental factor times the cumulative losses for that AY from the prior point.
2. Losses for a given accident year are independent of losses for any other AY.
3. Variance of incremental losses equals a constant times the incremental losses where the constant only depends on the development age.

b)

1. Assumption #2 above is violated. The incremental losses along the diagonal will be influenced and will affect all AY's violating the independence assumption.
2. This does not violate any assumptions and is needed for assumption #3 above to hold
3. This violates assumption #1 above. The incremental losses in 36-48 are dependent upon the losses from 24-36. For #1 to hold, they are only dependent on the loss at time 36, not what happened prior to 36 months.

Solution 2

a)

- 1) Expected increments in next period equals loss paid to date (in AY) times a factor based on age.
- 2) Variance is proportioned to losses paid to date (times a factor α_k^2 based on age)
- 3) Losses not in the same accident year are independent on each other.

b)

- 1) Yes, (c) would be violated. Calendar year effects mean that losses in various accident years have dependencies. Also, (a) may be violated; same factor based on age may not apply with calendar year effects.
- 2) None violated. Mack uses this too. Variance assumption depends on it.

- 3) Yes, if higher dev factor follows a lower one and vice versa, this implies that (a) is violated. The factors based on age are not randomly varying around some true factor, so we don't want to use a model that assumes one true underlying factor based on age to apply to cumulative losses.

Solution 3

a)

- 1) The cumulative claims in a period only depend on the previous level of cumulative claims and a factor based on age. $\sum(C_{i,K+1}) = C_{iK}f_K$
- 2) Accident years are independent of each other.
- 3) The variance of the next period estimates is proportional to the previous cumulative. $\text{Var}(C_{i,K+1}) = C_{i,K} \cdot \alpha_k^2 \leftarrow$ factor based on age.

b)

- 1) The second assumption is violated as all of the accident years are being affected by this new claims management software and cannot assume that the AYs are independent.
- 2) none are violated. The all year volume wtd approach follows from the variance assumption. This is the basic chain ladder.
- 3) If columns of development factors are not independent, this violates Mack's first assumption. The development in a period should only be dependent on the prior cumulative and a factor for that age, not a prior development factor. Since it seems like the development is dependent on the prior development factor, it appears to violate Mack's assumptions.

Solution 4

a)

- i. Accident year losses are independent of each other.
- ii. Expected incremental losses are based on cumulative losses for the accident period to date.
- iii. Variance of the accident period is proportional to losses (paid/reported) to date and age.

b)

- i. Violated. This shows that accident year losses are not independent as there are calendar year effects.
- ii. Appropriate. The volume weighted link ratios produce minimum variance in the Mack model.
- iii. Violated. This shows correlations between the columns.

Examiner Comment

We also accepted the formulas that underlie Mack's assumptions

Many candidates lost credit for not explaining why the assumptions were valid or violated.